Weatherbridge Weather Signals Observations and Everyday Decisions

Duke Kunshan University | INFOSCI 301 | Autumn 2026

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Abstract. Weatherbridge is an interactive visualization for comparing weather observations with prediction-market signals while making data status visible. The technical baseline combines NOAA LaGuardia daily maximum temperatures and archived Polymarket NYC Yes-token prices for 17-28 August 2026. A Mexico City NomadCast case uses deterministic simulated weather, market, activity, and place records. Users select a date and context, then inspect linked charts, a network, a schematic map, and distributions to distinguish observations, market prices, and simulations. The primary frontier is Open Science, Data Governance, and Responsible AI. The runnable prototype, source-documented NYC comparison, repository architecture, and explicit labels are current outputs. The next evaluation tests whether users can interpret the distinctions and state a supported comparison. No user-comprehension, decision, community, or SDG benefit is yet demonstrated.

Keywords: Weatherbridge, weather uncertainty, prediction-market signals, data provenance, usability evaluation, SDG 13.3

1 Project Frame and Contribution

1.1 Question, user, task, and local context

Question: How can interactive visualization help people whose activities depend on weather compare prediction-market temperature signals with weather observations while understanding the limits of that comparison?

Intended users are people planning weather-sensitive activities, including boat operators, households drying food, and outdoor business owners.

Task: choose a date and context; compare a market signal with a weather record; distinguish observations, prices, and simulated context; then state one supported takeaway and one unsupported conclusion. Jinxi Watertown motivates the question; Mexico City is an illustrative interaction case. A partner community and consultation route remain to be selected.

1.2 Evidence-supported claim and boundary

Implemented capability: linked views place archived market signals and weather observations beside explicit source and data-status labels; users can inspect differences and a simulated activity context. Whether users interpret these distinctions correctly remains the next evaluation question.

Boundary: the NYC comparison covers 12 days and the Mexico City baseline is simulated. The project has not established forecast accuracy, causal effects, better decisions, financial benefit, or a validated community need.

1.3 Project contribution

Design contribution: linked views explicitly distinguish weather observations, market prices, and interface-test simulation within one comparison task. The planned usability study tests whether labels and interactions support correct interpretation.

Primary frontier: Open Science, Data Governance, and Responsible AI. Table 3 also addresses Scientific Communication for Emerging Technologies and Human-Centered Spatiotemporal Intelligence. Provenance and linked views are implemented; effects on understanding and relevance remain evaluation questions.

Intended contribution: make weather information and uncertainty easier for non-specialists to access, understand, and compare.

SDG connection: Target 13.3 concerns climate education, awareness, and capacity. Weatherbridge explores a related interpretation question; a short-term temperature comparison does not demonstrate climate-change understanding or target progress.

Current output: a runnable prototype and source-documented NYC comparison; Mexico City demonstrates interaction with simulated context. Intended-user relevance and educational effects require evaluation.

Figure 1 Five representations serving one comparison task

Representation	Task served	Input or interaction	Data status and interpretation limit
Basic idiom	Read a dated temperature value and range.	Choose date and temperature bin.	NYC chart separates NOAA observation from market price.
Network idiom	Trace weather-to-activity associations.	Select weather or activity group.	Edges summarize simulated descriptive associations, not causes.
Spatial or spatiotemporal idiom	Locate context and inspect a shared date state.	Choose date, place, or map category.	Schematic Mexico City map; place and activity data are simulated.
Interactive view	Inspect provenance and change comparison context.	Select date, weather, group, activity, or place; reset.	Labels distinguish observations, market signals, and simulation; interpretation requires testing.
Advanced representation	Compare distributions and similar days.	Switch bins or similarity context.	Relative pricing and similarity are exploratory views. Peer comparison with separate charts is planned.

2 Design Capability

2.1 Design pipeline

Domain: weather-sensitive planning under uncertainty. Data and task: compare dated market prices, weather values, and activity context while retaining status.

Idioms: daily chart, shared timeline, network, schematic map, and selection panels. Algorithms: date normalization, temperature-bin preparation, missingness checks, simulated association summaries, and similarity ranking; these organize interface data but do not validate calibration or causality.

The five representations share one selected date and context rather than answering separate questions.

Table 1. End-to-end design pipeline.

Domain	Data and task	Idiom	Algorithm
Weather-sensitive planning under uncertainty	Compare date, market price, weather value, and activity context without collapsing status.	Daily chart, shared timeline, network, map, selection panels.	Date normalization; bin aggregation; missingness; association and similarity logic.

2.2 Four validation levels

Table 2 records current evidence, principal threat, and named next test; Appendix A preserves the Week 5 baseline and Appendix C gives the peer-task protocol.

Table 2. Four-level validation summary.

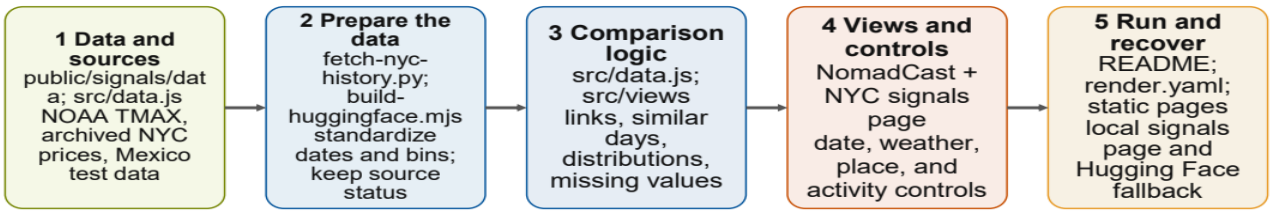
Level	Current evidence	Main threat	Next test and owner
Domain situation	Jinxi inspired the topic; intended-user hypothesis is explicit.	No Mexico City consultation or partnership.	Xinyuan: conduct exploratory peer tasks. Shudan: review the need narrative.
Data and task abstraction	NYC NOAA and archived Polymarket records are source-linked; Mexico data are labeled simulated.	Simulation or price may be mistaken for fact or accuracy.	Xinyuan: test source-status interpretation. Shudan: review labels. Yizhou: verify source definitions.
Visual encoding and interaction idiom	Linked controls and observed/inferred labels are implemented.	Linked views may overload or hide a state change.	Xinyuan: observe selection, recovery, and reset; retest after implementation.
Algorithm	Bins, association summaries, similarity, and hedge arithmetic are implemented.	No calibration, causality, liquidity, fee, or performance validation.	Yizhou: check transformations and missing values. Xinyuan: evaluate view comprehension.

2.3 Implementation, architecture, accessibility, and fallback

Stack: Vite and JavaScript modules, with static HTML/JS for NYC signals. Recorded technical baseline: main commit 2ae49b2, 27 September 2026.

The baseline records date and temperature-bin transformations and explicit missingness. Bilingual labels, direct source-status text, and reset controls are implemented; keyboard access and narrow-screen readability require evaluation. The next study compares the linked distribution view with separate charts. Archived Yes-token prices are signals, and normalized bins express relative pricing rather than calibrated weather probabilities.

Figure 2 Repository and software architecture



Architecture record	Data and provenance	Transform and store	Logic and algorithms	Views and interaction	Deploy, test, and fallback
Repository path or service	public/signals/data/; src/data.js	scripts/fetch-nyc-history.py; build-huggingface.mjs	src/data.js; src/views/; public/signals/src/	NomadCast map, network, panels; NYC signals page	README; render.yaml; local static fallback
Project-specific component	NOAA TMAX; archived NYC prices; deterministic Mexico test data	Normalize dates and bins; retain source status	Associations, similarity, distribution, hedge arithmetic	Date/weather/place/activity controls; bilingual labels	npm run dev/build; public/signals and Hugging Face fallback
Input, output, and check	Source records -> labeled data objects	Scripts regenerate NYC history only after checks	Logic returns explicit missingness and inferred states	Views display status text and direct boundaries	Open locally; reset controls; static file remains usable

Architecture text equivalent: NOAA observations and archived Polymarket prices feed the NYC comparison; deterministic Mexico City records feed NomadCast. Shared selections update the chart, map, network, and explanatory panels. Vite builds the application; static NYC and Hugging Face files are fallbacks.

Listed deployment URL: https://temperature-comparison.onrender.com/ . Static fallback: public/signals/index.html. The deployment address is supplied in the original poster; its live availability and deployed commit must be checked before submission.

Repository URL: https://github.com/Rick-1091/Temperature-Comparison

Technical baseline: 2ae49b2278f60759d38a35883a8c46a05273d832. The 6 October documentation revision changes wording and responsibilities only; it does not assert the live site runs this commit.

3 Interdisciplinary Contribution

3.1 Three emerging frontiers

Primary frontier: Open Science, Data Governance, and Responsible AI. Table 3 also addresses Scientific Communication for Emerging Technologies and Human-Centered Spatiotemporal Intelligence. Provenance and linked views are implemented; effects on understanding and relevance remain evaluation questions.

Table 3. Frontier-to-design contribution.

Emerging frontier	Contribution and evidence	Visible design consequence	Current limit
Open Science, Data Governance, and Responsible AI	README documents station, dates, source paths, raw price record, and data status.	Source explanations and simulation labels expose provenance for inspection.	Confirm API terms and license; add access date panel.
Scientific Communication for Emerging Technologies	Market prices are communicated as signals, with raw and normalized values distinguished.	Distribution, method note, and boundary are visible.	Xinyuan will test interpretation; Shudan will review terminology.
Human-Centered Spatiotemporal Intelligence	Linked date, weather, activity, and place controls supply a time-place context.	Schematic map and observed/inferred labels are visible.	Intended-user relevance and interpretation are unverified.

3.2 Research-to-design change

Data documentation changed the interface: Yes-token prices are not participant counts, bins can have different quote times, and prices need not sum to 100%; NOAA daily TMAX can differ from a market settlement definition. The interface therefore separates observations, raw prices, and relative distributions, and labels Mexico City records as simulated.

Source: repository README and public/signals/data/nyc-aug-2026.json, commit 2ae49b2.

4 Community-Rooted Global Leadership

4.1 Local community service

Audience: people whose daily activities depend on weather, including boat operators, households drying food, and outdoor business owners.

Jinxi observations motivate weather-sensitive work; the Mexico City module is a simulated interaction example, not evidence of community consultation.

Community engagement remains planned: no partner, consultation, partnership, or co-design outcome is documented. Xinyuan will evaluate usability with peer volunteers, not community representatives; Shudan will review narrative, terminology, captions, and source explanations. The public interface should offer accessible explanations, a feedback route, and dated corrections.

4.2 SDG goal and target

SDG 13, Target 13.3: Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning.

Intended educational contribution: help users distinguish weather observations, market signals, and simulated context. This connects the design to SDG 13.3 but is not evidence of climate-education or adaptation outcomes.

Current evidence is the documented technical baseline and source records; no user-evaluation finding or community outcome is supplied.

Limit: no awareness, adaptation, decision-quality, or SDG outcome has been measured.

Table 4. Local service and SDG accountability.

Accountability field	Current evidence	Limit or next check
Need hypothesis and missing voices	Initial audience hypothesis: people whose activities depend on weather may benefit from interpretable comparisons.	No consultation outcome is documented. Peer evaluation is assigned to Xinyuan.
Permission, protection, and public-sharing boundary	No personal or restricted data; Mexico records are deterministic test data.	Keep peer notes anonymized; verify source terms before public release.
Reciprocal return and correction route	Source links, static fallback, accessible text, and a future feedback route are planned.	Add visible feedback and dated correction log after consultation.
SDG 13.3 climate awareness and capacity	Prototype supports source-aware weather-uncertainty interpretation.	No impact measured; test interpretation before any SDG benefit claim.

5 Interactive Evaluation and Evidence-Based Revision

5.1 Activity 1 Interactive Demonstration

Baseline walkthrough: open NomadCast, select a future date and Outdoor cafe, inspect the simulated distribution, then select a historical date or Rainy context and inspect the linked map and network. Reset controls or public/signals/index.html provide fallback. No external breakdown is documented; labels and reset controls are safeguards to evaluate, not demonstrated learning outcomes.

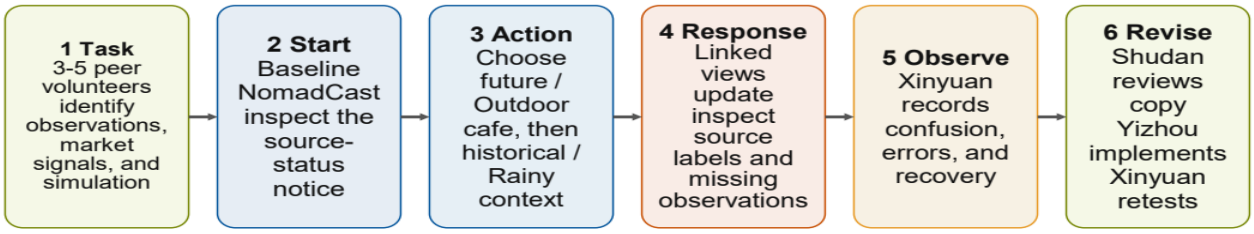
5.2 Activity 2 Peer Evaluation user study

The original peer study was scheduled for 29 September, but no results are supplied. Xinyuan is assigned to recruit 3-5 volunteer peers and record task paths, hesitations, interpretation errors, recovery, and language or keyboard barriers. Participants identify observations, market signals, and simulations and state one supported and one unsupported conclusion. Anonymized notes, a prioritized issue list, and a repeat check follow; this exploratory sample cannot establish broad usability or community need.

5.3 Activity 3 Response to Peers

After testing, Xinyuan will classify issues as Keep, Revise, or Investigate from observed behavior and severity. Shudan will propose bilingual content revisions with source checks; Yizhou will implement technical changes and record the commit. These are assigned next steps, not completed outcomes.

Figure 3 User walkthrough and evaluation plan



Study record	User and exact task	Starting state	User action	System response and task result	Observed evidence	Feedback decision and revision
Task path	Peer proxy: identify observed, market-implied, and simulated values for a selected date/context.	Default NomadCast state.	Select a future date and Outdoor cafe, then a past date and Rainy context.	Linked views update; status labels and boundary should remain visible.	External observation pending.	Investigate; record breakdown and revise.
Evidence boundary	Participant sees a prototype and data-status notice.	No personal data collected.	Actions and comments logged anonymously.	No forecast or causal conclusion requested.	No completed participant session supplied. Exploratory study assigned to Xinyuan.	Record actual findings after testing; content proposals by Shudan, implementation by Yizhou.

Walkthrough text equivalent: the participant changes a date or activity, inspects linked views, and identifies data status and a comparison limit. Xinyuan records confusion and recovery; Shudan reviews wording; Yizhou implements changes. The diagram records a planned process, not observed results.

5.4 What the evidence does not establish

The comparison is descriptive. NOAA daily maxima and market settlement records can use different definitions, so a match or mismatch does not establish settlement correctness or forecast accuracy. Relative pricing is not calibrated probability; simulated activity links do not establish causation. No study result, decision improvement, financial return, community benefit, or SDG outcome is documented.

6 Results, Limitations, and Next Validation

Table 5. Current result and next validation.

Claim or task	Current evidence	Main limitation	Next test and owner
Inspect a real dated comparison	Source-documented NOAA temperatures and archived Polymarket prices, 17-28 August 2026.	Small window; source and settlement definitions differ.	Verify provenance and explain mismatch. Yizhou.
Inspect a Mexico City decision context	Runnable linked-view prototype.	All Mexico case records are deterministic simulation.	Xinyuan: evaluate simulated-context comprehension. Yizhou: implement agreed technical changes.

7 Bounded Contribution and Current Conclusion

Weatherbridge makes weather observations and prediction-market signals inspectable through linked visualizations and explicit provenance. Jinxi motivates the everyday-use question; the Mexico City simulation supplies interaction context. Yizhou developed the system; Xinyuan is assigned evaluation and Shudan bilingual content review. Task evidence is required before claims of improved comprehension or practical benefit.

Acknowledgements

Acknowledgements. Yizhou Zhang developed and integrated the project. Xinyuan Lan and Shudan Guan are assigned the evaluation and content work in Appendix F. We credit NOAA NCEI, archived Polymarket records, and course materials. OpenAI Codex assisted drafting and wording revision; AI prose is not research evidence.

References

Project sources:

Munzner, T. (2009). A nested model for visualization design and validation. IEEE TVCG 15(6), 921-928. https://doi.org/10.1109/TVCG.2009.111

Dimara, E., and Perin, C. (2020). What is interaction for data visualization? IEEE TVCG 26(1), 119-129. https://doi.org/10.1109/TVCG.2019.2934283

Zhou, R., et al. (2023). Demonstrating TactorBots: A Haptic Design Toolkit for Exploration of Emotional Robotic Touch. CHI EA 2023, Article 438, 1-5. https://doi.org/10.1145/3544549.3583897

CHI 2026 Interactive Demos. https://chi2026.acm.org/authors/interactive-demos/

SIGCHI Guide for Videos. https://sigchi.org/resources/guides-for-authors/videos/

NOAA NCEI Daily Summaries, station USW00014732, TMAX, 17-28 August 2026; Polymarket event and price-history records documented in public/signals/data/nyc-aug-2026.json; Temperature Comparison repository, commit 2ae49b2; United Nations SDG 13 target 13.3, https://sdgs.un.org/goals/goal13.

Appendix A Evidence Map and Version History

This table separates the original Week 5 technical record from the 6 October documentation revision. Original planned dates do not establish completion. Later evaluation findings must be added with their actual session date and evidence.

Table A1. Evidence and revision history.

Stage	Version or evidence	Change and reason	Current location
Tuesday exit gate	Worksheet and role plan completed.	Scope clarified: Jinxi inspiration; NYC source-documented data; Mexico interface prototype.	Worksheet photos and repository history.
Interactive Demonstration	Internal 90-second walkthrough.	Added status labels and fallback path; external breakdown pending.	Current repository, commit 2ae49b2.
Week 5 first attempt	Current report and working repository.	Submitted baseline; planned items labeled with owners.	This report and repository main branch.
Peer Evaluation	Originally scheduled for 29 September; completion unconfirmed.	Test source-status comprehension and barriers.	Appendix C revised protocol; Xinyuan owns the assigned evaluation.
Response to Peers	Planned after peer evidence.	Keep, Revise, or Investigate each priority.	Appendix D: Xinyuan records findings, Shudan proposes wording, Yizhou implements changes.
Improved final attempt	Planned final resubmission.	Integrate actual evaluation evidence, revised poster, video, and release records when available.	Same Canvas thread and repository; all team.
Documentation revision	6 October 2026	Corrected wording, development attribution, status records, and assigned evaluation/content work.	Revised report and poster; technical baseline remains 2ae49b2.

Table A2. Four validation levels.

Level	Current evidence	Main threat	Next test and owner
Domain situation	Jinxi inspiration; Mexico use is a hypothesis.	No local consultation.	Xinyuan: peer tasks; Shudan: need narrative.
Data and task abstraction	NYC sources linked; Mexico simulation labeled.	Status confusion.	Xinyuan: status questions; Shudan: labels; Yizhou: sources.
Visual encoding and interaction idiom	Controls and labels implemented.	Interaction overload.	Xinyuan: observe and retest interactions.
Algorithm	Logic implemented; no accuracy/cause validation.	No calibration or performance validation.	Yizhou: transformation checks; Xinyuan: comprehension.

Appendix B Data Governance and Reproducibility

The data record distinguishes source-documented NYC observations and archived market prices from deterministic Mexico City simulation in the technical baseline. Each source has its own date, definition, provenance, and interpretation limit.

Table B1. Dataset and governance record.

Data record	Exact source and use	Risk or restriction	Verification
Primary dataset	NOAA NCEI Daily Summaries, USW00014732, TMAX, 17-28 Aug 2026.	TMAX may differ from market settlement definition.	Official API and repository snapshot; date and station recorded.
Complementary data	Archived Polymarket NYC daily high-temperature Yes-token prices and IDs.	Price times differ by bin; price total need not equal 100%.	Raw JSON, method, and market links in repository.
Synthetic or interface-test data	Baseline Mexico City weather, market, activity, and place records in src/data.js are deterministic simulation.	Not real Mexico observations, markets, or user behavior.	Code comment and interface copy label deterministic simulation.
Images, stories, or cultural material	Jinxi origin photos and story are contextual material only.	No claim of current community study or representation.	Repository jinxi/ assets; do not use as Mexico evidence.

Table B2. Reproducibility and release record.

Field	Record
Repository and artifact URLs	Repository: https://github.com/Rick-1091/Temperature-Comparison. Listed deployment: https://temperature-comparison.onrender.com/ . Static fallback: public/signals/index.html. Yizhou owns deployment verification.
Tagged version or commit	Technical baseline: 2ae49b2278f60759d38a35883a8c46a05273d832, 27 September 2026. Documentation revision: 6 October 2026. Live release identity requires confirmation.
Repository and software architecture figure	Figure 2 and the architecture table describe sources, preparation, logic, views, and fallback paths.
Environment and dependencies	Node, npm, Vite; JavaScript modules; Python polymarket-client and httpx for NYC refresh script.
Run and reproduction steps	Main Vite application: install dependencies with npm install, then run npm run dev; npm run build produces the static build. Static baseline fallback: open public/signals/index.html or huggingface/index.html. Do not assume the Vite entry runs by opening its HTML file directly.
Data access or generation steps	Run scripts/fetch-nyc-history.py only after required source checks; Mexico module is deterministic test data.
Tests and expected outputs	Open NYC comparison, select a date/bin, then open NomadCast and change date/context; labels should update.
Known issues and fallback	Deployment address is listed but live availability and commit are not confirmed here. Use the documented static fallback. Mexico City baseline activity records are simulated.
License, citations, privacy, and public-sharing decision	Cite NOAA, Polymarket, repository, and reused assets. Do not expose personal data; verify terms before public release.

Appendix C User Experience Protocol and Evidence

Xinyuan is assigned to conduct the exploratory peer evaluation. The protocol uses 3-5 volunteer peers and anonymized session records. Participants are peer testers rather than intended-user or community representatives. No completed sessions are reported here.

Table C1. User-experience session record.

Field	Record
Participant role and consent or notice	Assigned study: 3-5 volunteer peers, informed that the interface contains observations, market signals, and simulation. No participant identifiers retained.
Exact task and starting state	From default NomadCast state, identify observed, market-implied, and simulated values for selected context.
Success criterion and measures	Exploratory criterion: identify all three data statuses and state one supported and one unsupported conclusion. Record task completion, hesitation, errors, and recovery. Report counts with the actual participant denominator.
Actions, hesitation, errors, and recovery	Planned observation: selection path, hesitations, status confusion, reset use, and recovery.
System response, evidence inspected, and task outcome	Planned: linked chart/map/network update and status text; observer records outcome.
Accessibility or interpretation barrier	Planned: language, color/status-label recognition, keyboard path, and visual overload notes.
Observed strength, priority, and question	Planned question: does the user mistake market price for forecast accuracy or activity association for cause?
Feedback decision and visible revision	Xinyuan records issues and retests; Shudan proposes content changes; Yizhou implements technical changes. No completed findings supplied.
Evidence boundary and next validation	A small peer sample provides exploratory task evidence, not broad usability or community validation. Retest after revision.

Privacy and evidence handling

The observer will record screen state, task actions, optional verbal comments, outcome, and interpretation barriers. No name, contact information, or sensitive personal data will be collected. Team members may access the record for course work. Public materials will summarize feedback without identifying a participant.

Appendix D Feedback and Response Log

Table D1. Three-activity feedback and response record.

Activity or ID	Evidence received	Keep, Revise, or Investigate and reason	Change or next test and owner	Artifact or commit
Interactive Demonstration	Internal walkthrough completed; external observation pending.	Investigate because no external behavior is recorded.	Yizhou maintains the baseline walkthrough; Xinyuan observes the assigned peer sessions.	main 2ae49b2
Peer Evaluation	Originally planned for 29 September; completion unconfirmed.	Investigate source-status comprehension and barriers.	Xinyuan: conduct 3-5 exploratory peer sessions using Appendix C.	Planned commit
Response to Peers	Planned after peer evidence.	Keep, Revise, or Investigate based on observed evidence.	Xinyuan: prioritize and retest; Shudan: editorial proposals; Yizhou: code and commit.	Planned commit
Additional feedback	No additional feedback supplied in this revision.	Investigate until evidence exists.	Add only with source and owner.	Appendix D update

Feedback decisions will cite the observed task behavior and the reason for keeping, revising, or investigating an issue. Xinyuan owns evaluation records, Shudan owns editorial proposals, and Yizhou owns technical implementation. The record will link each completed change to its actual artifact or commit.

Appendix E Poster and Demo Evidence

Table E1. Poster preflight.

Check	Evidence or location
Descriptive title; evidence-supported claim and adjacent boundary	One-page revised poster PDF and editable PPTX accompany this report.
Question, users, exact task, source-documented data, and five representations	Report Sections 1 and 2; include NYC source-documented versus Mexico simulated distinction.
Repository and software architecture diagram from Figure 2	Figure 2 architecture illustration and table describe the technical baseline; technical owner: Yizhou.
User-experience walkthrough and revision diagram from Figure 3	Figure 3 illustrates the planned evaluation. Xinyuan owns observation; Shudan reviews wording; Yizhou implements changes.
Four validation levels and all three frontiers, with one primary	Report Tables 2 and 3. Primary: Open Science, Data Governance, and Responsible AI.
Local-community service and exact SDG goal and target	Report Section 4; Mexico use is hypothesis; SDG 13.3.
Provenance, written URLs, roles, accessibility, and text equivalent	Appendices B and F distinguish source status and responsibilities. Deployment address is listed; live version, accessibility checks, and member review remain to confirm.

Table E2. Demo storyboard and transcript map. Total duration must be 2:59 or shorter.

Time	Video segment	Required content and evidence
0:00-0:15	Opening	Question, intended user, exact task, local need hypothesis, and claim boundary
0:15-0:45	System architecture	Repository and software architecture, submitted version, data flow, and fallback
0:45-1:45	User walkthrough	Starting state, interaction, system response, task result, and evidence inspected
1:45-2:20	User evidence and revision	If sessions have been completed, show actual behavior, decisions, and changes. Otherwise identify the evaluation plan and pending evidence explicitly.
2:20-2:50	Contribution and next validation	Primary frontier, other two frontiers, local service, SDG target, limitation, and next test
2:50-2:59	End card	Project title, team, written artifact and repository URLs

Table E3. Professional reference artifacts.

Reference	What to study	Course extension
CHI 2026 Interactive Demos: https://chi2026.acm.org/authors/interactive-demos/	Self-contained demo paper, walkthrough video, accessible submission, and successful examples.	Include architecture, user walkthrough, frontier connections, and exact SDG target. Record peer evidence only after collection.
Zhou et al. TactorBots demo paper: https://doi.org/10.1145/3544549.3583897	How a short demo paper explains a working system and the interaction experience.	Make repository structure, data provenance, evidence boundaries, and correction routes explicit.
TactorBots video: https://www.youtube.com/watch?v=mGvSld62Mhk	How the system, user flow, and physical or interface behavior become visible on video.	Keep the course video to 2:59 or shorter and show actual evaluation evidence when available. Otherwise explain the pending study.
Zhou et al. Shape-Kit demo paper: https://doi.org/10.1145/3706599.3721280	How a recent CHI demonstration connects system design, interaction, and a concise research contribution.	Add the repository architecture, the task-based walkthrough, and explicit evidence boundaries.

Reference	What to study	Course extension
Shape-Kit video: https://www.youtube.com/watch?v=JSUDrGJjpWw	How a working system and user experience can be communicated through a short demonstration.	Use the stricter course duration and show the feedback-to-revision trace.
CHI 2026 Posters: https://chi2026.acm.org/authors/posters/	Title, concise overview, clear illustrations, visual legibility, and links to online materials.	Show software architecture, UX walkthrough, three capacities, local community, and SDG accountability.
AAAI-26 Student Abstract and Poster Program: https://aaai.org/conference/aaai/aaai-26/student-abstract-and-poster-program-call/	Concise paper-to-poster translation in a major AI conference context.	Preserve the visualization validation and public-service requirements of this course.
SIGCHI Guide for Videos: https://sigchi.org/resources/guides-for-authors/videos/	Captions, audio, readability, permissions, encoding, and accessible visual description.	Follow the course's 2:59 maximum and verify every caption and technical term.

Appendix F Team Contribution and AI Assistance

Table F1 Completed development and assigned next work

Member	Completed work or assigned responsibility	Evidence and status	Scope and next deliverable
Yizhou Zhang	Technical Architecture and System Development Lead. Developed the full system, including data acquisition and processing, visualization, interaction logic, integration, deployment, and technical verification.	Development attribution corrected by Yizhou on 6 October 2026. Technical baseline: commit 2ae49b2. Detailed commit-by-commit attribution is not audited in this revision.	Own system architecture, sources and transformations, all coded views and controls, reproducibility, and deployment. Implement agreed changes and record technical validation.
Xinyuan Lan	User Experience Evaluation and Usability Improvement Lead. Assigned next work: evaluate navigation, chart comprehension, data-status interpretation, and recovery; prioritize issues and conduct follow-up checks.	New responsibility assignment, 6 October 2026. No completed user sessions or evaluation findings supplied.	Deliver anonymized records from 3-5 volunteer peers, a prioritized issue list with page/task evidence, and a repeat check after changes. Peer findings do not establish community impact.
Shudan Guan	Content Strategy and Bilingual Communication Lead. Assigned next work: organize narrative flow, review Chinese and English copy, and align terminology, captions, source descriptions, and project materials.	New responsibility assignment, 6 October 2026. Editorial completion will be documented after delivery and review.	Deliver a before/after copy-edit table with reasons, a website/report/poster consistency checklist, and a demo script. Yizhou implements changes requiring code.

AI assisted report drafting and the 6 October editorial revision. The development attribution follows Yizhou's correction. The evaluation and content roles are new assignments, not retrospective development credits. Final member review remains a submission step.

Table F2. AI assistance and human verification.

Field	Disclosure
Tool, displayed model or version, and date	OpenAI Codex. Original draft dated 27 September 2026; documentation revision dated 6 October 2026. The original displayed model version is not confirmed.
Purpose and significant prompt or input summary	Original drafting from the course materials and repository record; revision of wording, evidence boundaries, status records, and team responsibilities following Yizhou's corrections.
Output accepted and where it appears	Revised report prose, tables, evaluation plan, and poster wording. Final review by each member remains required before submission.

Field	Disclosure
Output rejected and why	This revision excludes unverified completed tests, retrospective development credit for assigned nontechnical work, and claims of proven accuracy, causal impact, or financial benefit.
Human changes to argument, visuals, data, or code	Yizhou supplied corrected development attribution and approved evaluation/content assignments. Changes distinguish implemented capabilities from planned work and measured outcomes.
Original sources, data, code, and outputs checked	Original report and poster were inspected. The project package scripts were checked for Vite run/build commands. Live deployment, original model identity, test completion, and individual member approvals are not confirmed.
Failures corrected and remaining unverified points	Corrected role attribution, inconsistent community wording, old planned-date phrasing, and poster/deployment status. Remaining checks: live release identity, evaluation outcomes, video, licenses/source terms, and final member review.

Team responsibility statement

Responsibility record: Yizhou is responsible for the technical implementation and data processing. Xinyuan is assigned usability evaluation and follow-up checks. Shudan is assigned content organization, bilingual review, and project communication. Each member will verify their own completed work and its evidence before submission. The report distinguishes implementation, simulation, planned evaluation, and measured outcomes.

Members listed for review: Shudan Guan, Yizhou Zhang, Xinyuan Lan. Documentation revision: 6 October 2026. No new signatures or individual approvals are asserted.